

Plotting, Ordering and Comparing Real Numbers

Additional Practice 7A.02.06

Name _____

Date _____

Period _____

Optional practice. *Same four parts, different numbers.***PART 1. Comparing***Use $<$, $>$ or $=$. One of the six is an equality.*

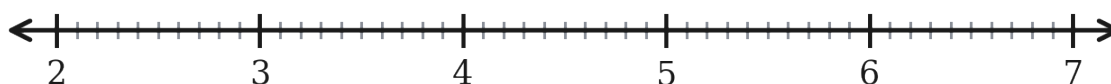
1.	$-2\pi + 10$		$\sqrt[3]{-56}$
2.	$\sqrt{300}$		$\frac{196}{11}$
3.	$-\sqrt{\frac{162}{2}}$		$-9.\overline{09}$
4.	$3\sqrt{100}$		π^3
5.	$\sqrt{144}$		12
6.	$\sqrt[3]{65}$		$\frac{17}{4}$

PART 2. Sorting and plotting*A number is irrational only if it cannot be written as a ratio of two integers.*

7. Sort the five expressions, then plot and label each one on the number line.

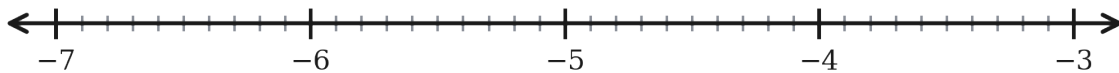
$$2\frac{1}{2} \quad \sqrt{10} \quad \sqrt[3]{64} \quad \pi + 2 \quad \frac{19}{3}$$

Only rational numbers	Includes an irrational number



8. Plot and label each expression on the number line.

$$-\sqrt{45} \quad -\frac{19}{3} \quad -\sqrt{30} \quad -\sqrt{27} \quad -\sqrt[3]{64} \quad -\sqrt{14}$$



PART 3. Error analysis

Read the work, find the error, then do it correctly.

9. Devin approximated $\sqrt{40}$ like this:

“40 is between the perfect cubes 27 and 64, so $\sqrt{40}$ lies between 3 and 4.”

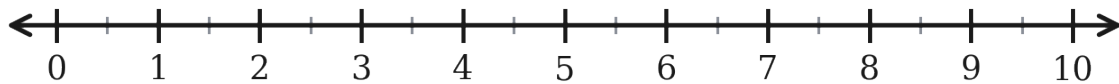
- a. What is Devin's error?

- b. Redo it correctly.

PART 4. Ordering and plotting

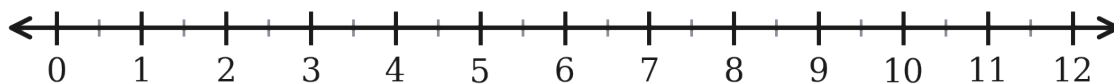
10. Order the numbers from least to greatest, then plot and label each one.

$$\sqrt[3]{1} \quad \sqrt[3]{27} \quad \sqrt{15} \quad \sqrt{40} \quad \frac{31}{4} \quad \pi^2$$



11. Order the numbers from least to greatest, then plot and label each one.

$$\frac{1}{2} \quad \sqrt{8} \quad 5\frac{1}{2} \quad \sqrt{55} \quad 3\pi \quad \sqrt{126}$$



STRETCH. *Beyond the range the test uses. Same method, bigger cubes.*

12. Find the value of $\sqrt[3]{512}$ and of $\sqrt[3]{-216}$.